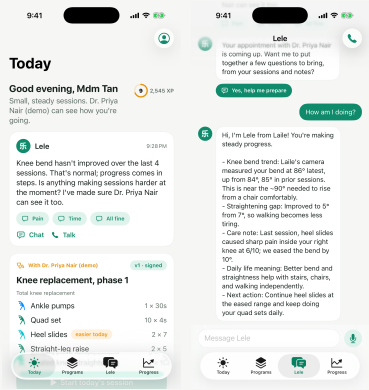
# Laile

Home physio that counts every rep and checks in.

Tencent Cloud AI CAN DO IT Hackathon Singapore 2026
Healthcare track • Challenge Statement 2

Charles Andrew Lim



# Physio happens at home, where nobody sees it

**50%**

of people in rehab may not do their home exercise [1]

**21.4%**

of Singapore citizens are 65 or older (2026), a super-aged society [3]

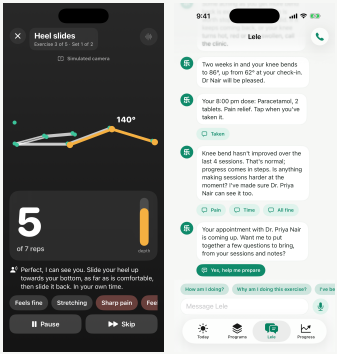
**Why people stop** (systematic review of 20 studies [2]): low confidence, low mood, worry, pain during exercise, no support, too many barriers.

**What a physio sees between visits:** nothing. A checkbox can't tell a done exercise from a skipped one, or one stopped because it hurt.

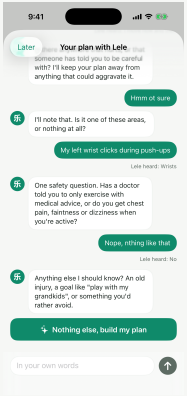
**What the brief asks for:** support that is empowering, personalised, longitudinal, safe, strictly bounded and honest about uncertainty.

# Laile in one breath

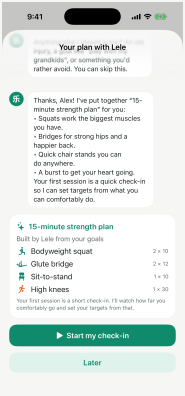
- **Programs are the unit.** A physio's program (joined with their code), a plan Lele builds from your goals, or a calisthenics program.
- **The camera counts.** Reps, holds, knee bend and arm lift, measured on the phone. Video stays there.
- **Lele checks in.** A coach that talks you through sessions, asks how you are each morning, reminds, notices, remembers what hurt.
- **The physio stays on top.** Live sessions, a daily list, approvals, signatures, discharge.
- **Code decides, the model talks.** Safety and every number come from tested rules.



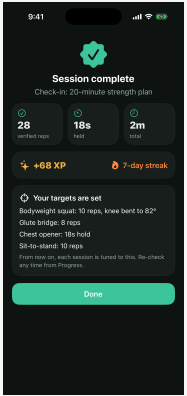
# Start: tell Lele your goals, or bring a physio's code



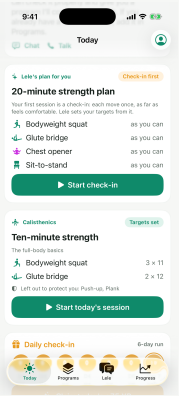

Goals in your own words; options are suggestions




Lele's plan leaves out what your wrists can't take

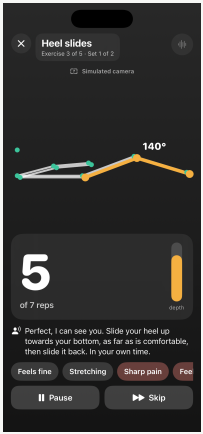



First session is a check-in that sets targets




Limits follow you into every program

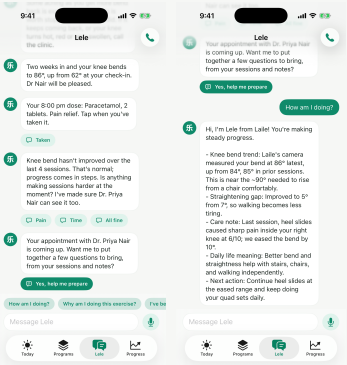
# Sessions: the camera counts, Lele coaches



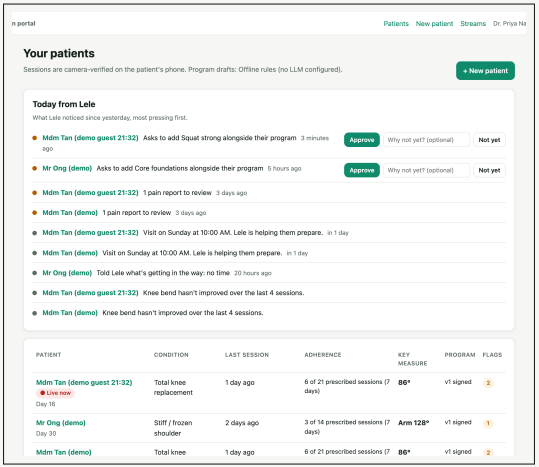

- Apple Vision tracks joints on the phone; a rep counts only at the target angle, a hold only while in position.
- Lele takes turns like a conversation: explains, says *"In your own time"*, counts, asks *"How did that feel?"* between sets.
- Say "it's pulling" or "sharp pain" any time. Fixed rules decide: carry on, stop the set, stop, or get help.
- Something sore? Lele notes it and makes that exercise easier next time, then eases back after two comfortable sessions.
- Can't see you? No count: *"I can't see you clearly, so I'm not counting that."*

# Lele takes initiative, and you can talk back

- **Morning check-in** in physio. “Quite swollen this morning” makes today’s session gentler and tells the physio.
- **Medicine reminders** at the doctor’s times, as notifications too; “What’s this for?” on every dose.
- **Missed days:** one nudge, then “what’s getting in the way?”
- **Plateaus and visit prep** from real measurements.
- **Chat or call** Lele, built on Tencent Cloud ADP with a curated knowledge base.
- **Never nags:** quiet 9 pm to 7 am, three conversations a day at most.

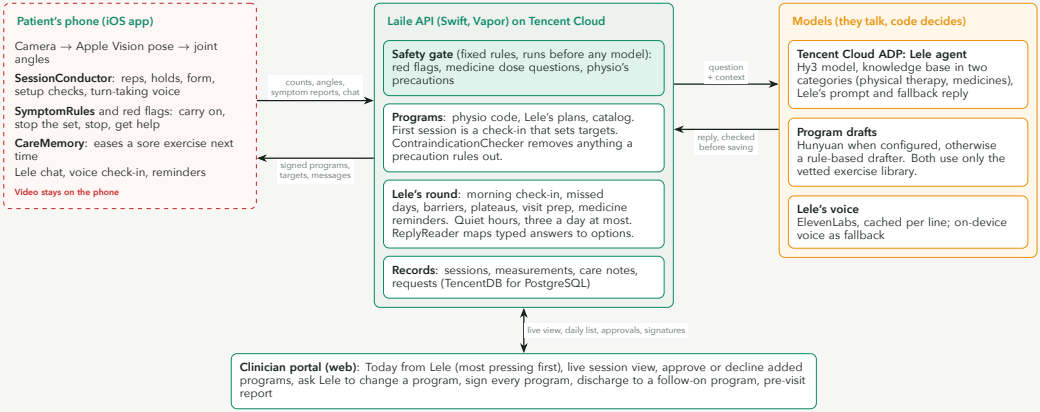


# The physio stays on top




- **Today from Lele**, most pressing first: red flags, worse this morning, pain reports, requests, missed days.
- **Live view** of a session: exercise, reps, measured angle.
- **Approve** extra programs; **ask Lele** to change one in plain words; **sign** every version.
- **Discharge** hands the patient to a follow-on program, limits kept.

# Architecture and trust boundary



# How the AI is used, and kept safe

| Job | Done by | Kept safe by |
|---|---|---|
| Conversation and explanations | Lele on **Tencent Cloud ADP** (Hy3, 12-document knowledge base) | Real context sent with each message; red flags and dose questions answered by rules first |
| Remembering what hurt | Care notes | Easing computed by code; physio sees every note |
| Program drafts | Hunyuan or rule-based drafter | Vetted library only, precautions enforced, physio signs |
| Understanding replies | Keyword rules | "Lele heard: …"; asks again if unclear |
| Seeing movement | Apple Vision on device | Video stays on the phone |

- Dose questions: fixed referral to the doctor or pharmacist. Unknown medicine: *"I'm not able to identify that medication, please check with your pharmacist."*
- 146 automated tests cover counting, safety rules, care notes, Lele's round and the server flows.

# Built with

### Tencent Cloud and Tencent tools

**ADP**: Lele agent on the Hy3 model, 12-document knowledge base, published chat link

**Lighthouse**: hosts the API and clinician portal (Docker, Caddy, HTTPS)

**Hunyuan**: program drafts when configured

**CodeBuddy / WorkBuddy**: safety and ADP audits, safety fixes (Hy4)

**Miora**: logo, cover, app icon and the Lele mascot (Hy Image 3.5)

### iPhone app

**Swift 6, SwiftUI**: feature modules in LaileKit

**Apple Vision**: body pose on the phone for reps, holds and joint angles

**AVFoundation, Apple Speech**: camera, spoken symptoms

**ElevenLabs**: Lele's voice, cached lines and live turns; on-device voice as fallback

**TestFlight**: iPhone builds for testers

### Server and shared code

**Vapor (Swift)**: API, Leaf clinician portal, live session view over WebSocket

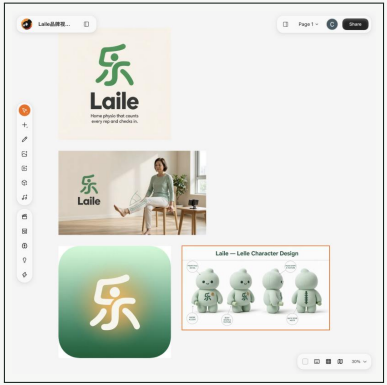
**Fluent**: SQLite, or TencentDB for PostgreSQL

**LaileCore**: one Swift package for safety rules, programs, care notes and Lele's round, shared by phone and server

**146 automated tests**: 117 core, 29 server

**Tooling**: XcodeGen, Docker Compose, Tectonic

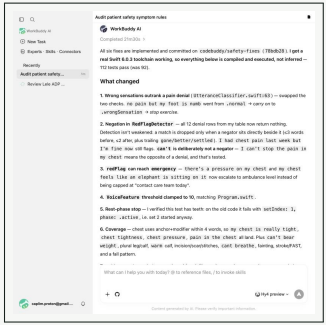
# Made with Tencent's own tools




**Miora**: logo, cover, app icon, Lele mascot




**WorkBuddy**: safety audit with file and line findings




**WorkBuddy**: six safety fixes, 112 tests passing

- **CodeBuddy** found that "no pain but my foot is numb" was answered with "carry on" and that "I haven't taken it yet" was logged as a dose; its fixes were reviewed and merged.
- **ADP** runs Lele; **Lighthouse** hosts the API and portal; **ElevenLabs** gives Lele a voice.

# Fit with Challenge 2

| | |
|---|---|
| Medication understanding | “What’s this for?” on every dose, from the medicines knowledge base |
| Medication management | Reminders at the doctor’s times; one tap or a typed reply marks a dose taken |
| Adherence support | Camera-verified sessions, check-in targets, forgiving streaks, barrier questions |
| Health tracking | Knee bend, straightening gap, arm lift, reps, holds and pain against baseline |
| Preventive health | Precautions follow you; plateau and missed-day flags; follow-on programs |
| Healthcare preparation | Visit prep questions; pre-visit report for the physio |
| Safe, bounded, honest | Rule-based red flags and referrals; no count without a clear view |

# Why it can last

**Clinics**
Verified adherence, live sessions and a prioritised daily list without extra visits. Proposed: a monthly fee per active patient.

**People**
Discharge leads to a calisthenics or mobility program with limits carried over. Consumer subscription for people without a physio.

**A loop**
Everyday users with aches are pointed to a physio; physio patients become everyday users afterwards.

**Cheap to run**
Pose estimation on the phone; model calls only for conversation; voice lines cached and paid for once.

# Where it stands

## Working now

- iOS app: Today, Programs, Lele, Progress
- Camera pipeline (Simulator, simulated skeleton)
- Server and portal live on Tencent Cloud Lighthouse
- Lele published on Tencent Cloud ADP
- Demo accounts with realistic history
- 146 automated tests

## Next

- TencentDB for PostgreSQL
- Real-iPhone testing; TestFlight for outside testers
- Knowledge base review by a physiotherapist and a pharmacist
- Pilot with one clinic

Source: github.com/LimCharles/laile • Try Lele: adp.tencentcloud.com/webim_exp/#/chat/nZbZZz
Portal: 43-156-108-79.sslip.io/portal (clinician@laile.demo / laile-demo-2026)
iPhone app: testflight.apple.com/join/WuymKzUH

# References


[1] Argent R, Daly A, Caulfield B. Patient involvement with home-based exercise programs: can connected health interventions influence adherence? *JMIR mHealth uHealth* 2018;6(3):e47.

[2] Jack K, McLean SM, Moffett JK, Gardiner E. Barriers to treatment adherence in physiotherapy outpatient clinics: a systematic review. *Manual Therapy* 2010;15(3):220-228.

[3] National Population and Talent Division. *Population in Brief 2026*. Singapore, September 2026.